

Zeel Gajjar

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Cybersecurity student exploring quantum computing and cryptography, with hands-on project and research experience in QKD, post-quantum cryptography, and quantum-inspired systems.

EDUCATION

MBIT, CVM University — Gujarat, India
B.Tech in Cybersecurity | Expected 2029

EXPERIENCE

Research Intern — QKD & RSA | Jun 2026 — Jul 2026

TechForGood 2026, IEEE MIT-ADT Student Branch — Remote

- Researched Quantum Key Distribution (QKD) protocols and RSA systems under IEEE Maharashtra Section.
- Investigated transitions to post-quantum standards, including lattice-based and hash-based schemes.
- Evaluated protocol resilience against emerging quantum threats through simulation and comparative security analysis of classical vs. post-quantum key exchange.

RESEARCH & PRESENTATIONS

DAU Silver Jubilee Seminar (ETSIF), Decoherence Effects in VQE of Phi-4 Scalar Field Theory | 2026

- Implemented VQE for 4-site lattice Phi-4 Hamiltonian using Qiskit; achieved 0.87% accuracy vs. exact diagonalization.
- Characterized noise effects (depolarizing, damping) and demonstrated robustness across multiple ansatz depths.
- Findings offer insight into how NISQ-era hardware noise limits variational algorithms simulating lattice field theories.

PDEU — Indo Quantum Summit 2026, Quantum-Inspired Random Number Generator | 2026

- Designed entropy harvesting combining photon shot noise and thermal noise; achieved 98.3% NIST SP 800-22 pass rate.
- Built for use in cryptographic key generation where high-quality randomness is critical to security.

SELECTED PROJECTS

Quantum-Inspired Behavioral Authentication

- Designed continuous authentication encoding dynamics as quantum state vectors; implementing CRYSTALS-Kyber.
- Uses STFT-based feature encoding and fidelity matching to flag anomalous behavior in real time, reducing reliance on static passwords.

Cryptographic Key Statistical Test Suite

- Implemented 34+ tests (NIST SP 800-22 battery + advanced tests) using Python/NumPy/SciPy.
- Automated pipeline benchmarks RNG/PRNG output quality, supporting validation for the entropy-harvesting RNG project.

Topological Characterization of Access Structures in Secret Sharing

- Analyzed access structures in secret sharing schemes (e.g. Shamir's Secret Sharing) through simplicial homology.
- Applied algebraic topology to characterize structural properties of authorized coalitions in threshold-based schemes.

LEADERSHIP & ACHIEVEMENTS

- Vice Chairperson: IEEE Signal Processing Society, MBIT.
- 1st Place: State-Level Ideathon (AI-integrated firewall with cryptographic anomaly detection).
- 1st Place: College-Level Ideathon (IoT encryption system using camera-based entropy extraction).
- CTF Competitor: Active in CryptoHack and OverTheWire (Web, Binary, Cryptography).

TECHNICAL SKILLS

- **Languages:** Python, C/C++, Julia, Bash, HTML/CSS.
- **Cryptography:** NIST testing, QKD (BB84), RSA, Post-Quantum (Kyber, SPHINCS+), Cryptanalysis.
- **Quantum & Signal:** Qiskit, VQE, NISQ noise simulation, FFT/STFT, EEG analysis.
- **Security Tools:** Burp Suite, Wireshark, Nmap, OpenCV, Vercel.